Asymptotic Notation Practice

Q1

- Prove that (2n2 + 3n + 4) is in O(n2)

$2n^2 + 3n + 4 \leq cn^2$

$c = 2 + 3 + 4 = 9$

$n_0 = 1$

Q2

- Prove that $100n + 10000$ is in $O(n^2)$

$100n + 10000 \leq cn^2$

$c = 10000 + 100 = 10100$

$n_0 = 1$

Q3

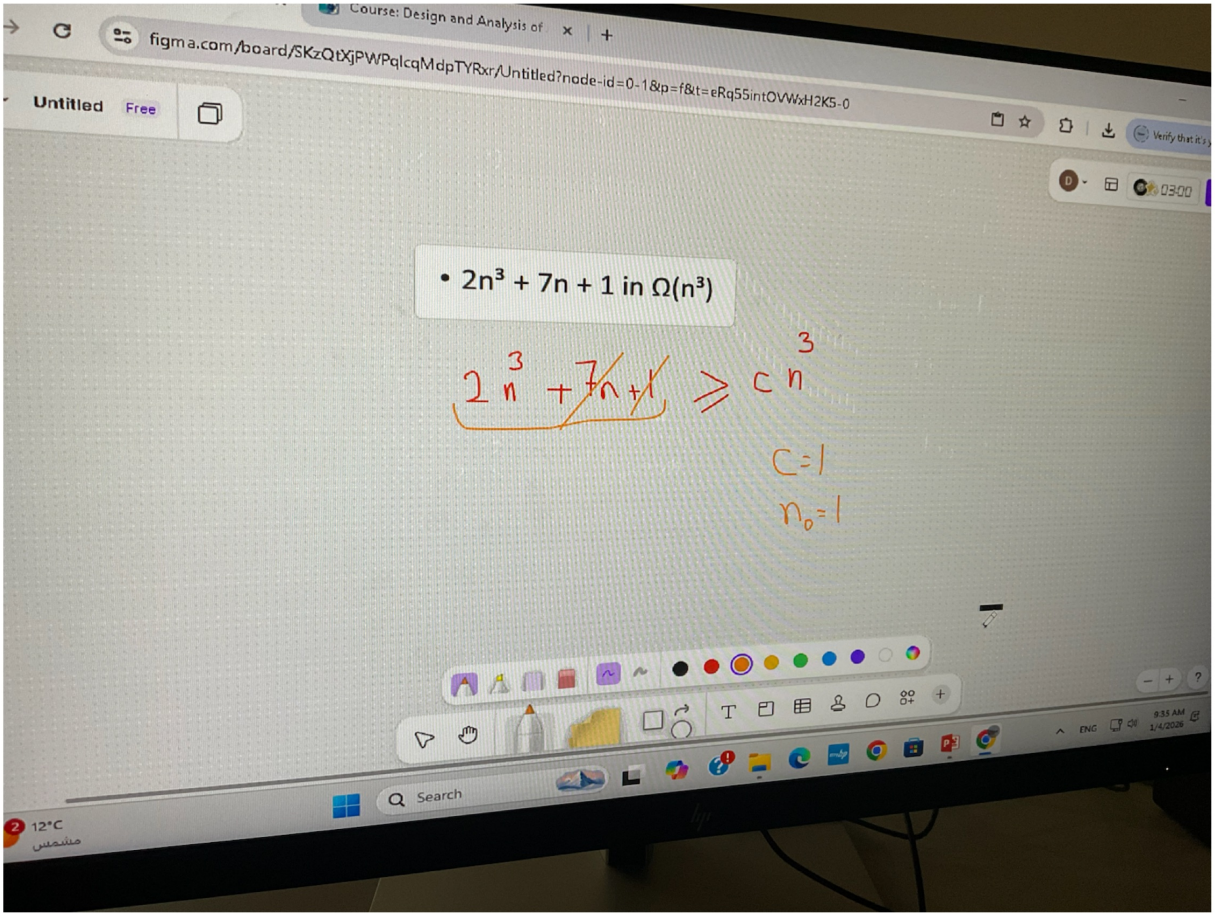
- Prove that $5n^2 - 3n + 20$ is in $O(n^2)$

ignore the negative since it'll make the number smaller

$5n^2 - 3n + 20 \leq cn^2$

$c = 5 + 20 = 25$

$n_0 = 1$


Course: Design and Analysis of
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• 2n³ + 7n + 1 in Ω(n³)
2n³ + 7n + 1 ≥ cn³
c=1
n₀=1
Search
12°C

Q4

- Prove that $2n^3 - 7n + 1$ in $\Omega(n^3)$

Q5

- Prove that $2n^3 - 7n + 1$ in $\Theta(n^3)$

Q6

- Show that $\sqrt{6n^3 - 7n^2 + 3n +}$

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a) Rank the following time complexity from the fastest to slowest (2 pts)
O(nlog n), O(2^n), O(n^2), O(1), O(log n), O($\sqrt{n}$), O(n!)

fastest

O(1)

log (n)

$\sqrt{n}$

n log n

n^2

n

2^n

n!

slowest

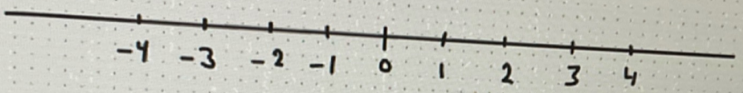
$2^n = 2 \times 2 \times 2 \times \cdots \times 2$ (n times)

$n! = 1 \times 2 \times 3 \times \cdots \times n$

$n^n = n \times n \times n \times \cdots \times n$

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Smaller ← → larger



True or False:

$2 < 4$ T

$2 < 2^2$ T

$n < n^2$ T

$n < n^3$ T

$-2 > -4$ T

$-2 > -2^2$ T

$-n > -n^2$ T

$-n > -n^3$ T